

Signatures of shear thinning-thickening transition in steady shear flows of dense non-Brownian yield stress systems

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Steady shear flows of dense athermal systems composed of soft disks are investigated via non-equilibrium molecular dynamics simulations, from which we sort out links among the structure, dynamics, and shear rheology. The systems at rest are jammed packings of frictionless disks with a nonzero yield stress. Driven by low shear rates, the flows shear thin due to the presence of the nonzero yield stress, but transit to shear thickening above a crossover shear rate $\dot{\gamma}_c$. At $\dot{\gamma}_c$, we observe the strongest structural anisotropy in the pair distribution function, which serves as the structural signature of the shear thinning-thickening transition. We also observe dynamical signatures associated with the transition: At $\dot{\gamma}_c$, scaling behaviors of both the mean squared displacement and relaxation time undergo apparent changes. By performing a simple energy analysis, we reveal an underlying condition for the shear thickening to occur: $d(\ln T_g)/d(\ln \dot{\gamma}) > 2$ with T_g the kinetic temperature. This condition is confirmed by simulations.

shear thickening, shear thinning, structural anisotropy, dynamics, kinetic temperature

1 Introduction

Complex fluids such as colloids, gels, emulsions, foams, and granular materials exhibit intriguing and complicated rheological phenomena when they are subject to shear. Unlike Newtonian fluids, whose shear viscosity is independent of the shear rate, complex fluids can undergo shear thinning or thickening under proper conditions; these are characterized by the decrease or increase of the viscosity as the shear rate increases [1–5]. Shear thinning and thickening are attractive research topics with important applications [2,6]. In practice, we always wish paints to shear thin, whereas shear thickening is desired in the design and manufacture of smart materials such as soft body armor.

The underlying mechanisms of shear thinning and thick-

ening have been debated for decades. Early simulations suggested that the layering of particles along the direction of the shear flow contributes to shear thinning [7,8]; such results have been suspected as artificial of profile-biased thermostats [9–12] and have been challenged by most recent studies [13,14]. A recent measure of microscopic single-particle dynamics has proposed that shear thinning results from the decrease of the entropic force contribution [3]. Shear thinning can also be boosted by the presence of yield stress which impedes the emergence of shear thickening [15].

Our understanding of shear thickening is even poorer than our understanding of shear thinning; this is largely because fewer systems exhibit shear thickening, which usually happens at high shear rates where probing is difficult. Shear thickening has been observed in both Brownian and non-Brownian suspensions [3,15,16–18] and in simulations with and without taking hydrodynamics into account [2,8,

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10–12,19]. Multiple mechanisms (e.g., the order-disorder transition and formation of hydroclusters) [2,8,20] have been proposed to explain shear thickening. These mechanisms can correctly predict the critical shear stress or shear rate of the onset of shear thickening. However, they fail to explain the occurrence of discontinuous shear thickening (DST), which is an extreme shear thickening phenomenon in which the shear stress jumps discontinuously with the shear rate; it usually happens at high packing fractions [5]. Recent studies that have unveiled a possible link between DST and jamming of constituent particles due to dilation [5,16–18,21–23] have proposed friction as the necessary element [24,25]. All of these explanations, which are based on different physics, have their strengths and weaknesses. Whether shear thickening can or cannot originate from a universal mechanism for various systems remains a set of open questions.

We studied the rheology of planar shear flows of dense non-Brownian systems without the interference of hydrodynamics via non-equilibrium molecular dynamics simulations. The system at rest is a jammed packing of frictionless spheres with a nonzero yield stress. Due to the presence of the yield stress, the system shear thins instead of being Newtonian at low shear rates [15]. Above a crossover shear rate $\dot{\gamma}_c$ the system shear thickens. The existence of $\dot{\gamma}_c$ enables us to explore the structural and dynamical signatures associated with the shear thinning-thickening transition; to our knowledge, information about these signatures is still lacking in the literature. We observed that the shear thinning-thickening transition was accompanied by the strongest structural anisotropy and apparent changes in the scaling of dynamical quantities. Furthermore, we found that in order for the shear thickening to occur the increase rate of the kinetic temperature T_g with respect to the shear rate $\dot{\gamma}_c$, $d(\ln T_g)/d(\ln \dot{\gamma})$, must exceed 2, based on the energy balance between input and dissipation. This condition was confirmed by simulations and was not limited to the model system discussed here.

2 Simulation details

Our systems are two-dimensional $L \times L$ squares consisting of disks with an identical mass m . Half of the disks have a diameter σ , the other half have $\sigma_L = 1.4\sigma$. Lees-Edwards boundary conditions [26] were applied with the shear being imposed in the x direction and shear gradient in the y direction. SLLOD equations of motion assuming a linear velocity profile [27] were employed:

$$\frac{d\vec{r}_i}{dt} = \vec{v}_i + \dot{\gamma} y_i \hat{x} \quad (1)$$

$$\frac{d\vec{v}_i}{dt} = \frac{1}{m} \sum_j \vec{F}_{ij} - \dot{\gamma} v_{yi} \hat{x} \quad (2)$$

where $\vec{r}_i = (x_i, y_i)$ and $\vec{v}_i = (v_{xi}, v_{yi})$ are the location and random velocity of particle i , $\dot{\gamma} = d\gamma/dt$ is the shear rate with γ the shear strain, $\vec{F}_{ij}$ is the force acting on particle i by particle j , and the sum is overall particles j interacting with particle i . The force $\vec{F}_{ij}$ includes two parts, the elastic force $\vec{F}_{ij}^e = -\nabla V_{ij}$ and damping force $\vec{F}_{ij}^v = -\xi m (\vec{v}_{ij} + \dot{\gamma} y_{ij} \hat{x})$ [28], where V_{ij} , $\vec{v}_{ij}$, and y_{ij} are the interaction potential, relative random velocity, and y distance between particles i and j that are in contact with each other, and ξ is the damping coefficient. Weeks-Chandler-Anderson (WCA) potential (i.e., repulsive Lennard-Jones potential) is applied:

$$V_{ij} = \frac{\epsilon}{72} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 + 1 \right] \quad \text{when the separation be-}$$

tween particles i and j , r_{ij} , is smaller than the sum of their radii σ_{ij} , and zero otherwise. We used σ , m , and ϵ as the units of length, mass, and energy. The units of temperature and time are ϵ/k_B and $\sqrt{m\sigma^2/\epsilon}$, respectively, where k_B is the Boltzmann constant.

We applied Gear predictor-corrector algorithm to integrate Eqs. (1) and (2) at a constant packing fraction $\phi = 0.85$ above the $T = 0$ jamming transition at $\phi_c \approx 0.84$ [29–32]. The systems at rest are jammed solids obtained from L-BFGS energy minimization (<http://www.ece.northwestern.edu/~nocedal/lbfgs.html>). Data were collected and averaged over time after the systems were sheared for a much longer time than the characteristic time scale $1/\dot{\gamma}$ and steady shear flows without the memory of their initial states were achieved. We guarantee that the observation time was sufficiently long, so that the statistics are good enough and the results do not change quantitatively even when a longer time is applied. We have also verified that our results do not depend on initial conditions. The shear stress Σ_{xy} is calculated from

$$\begin{aligned} \Sigma_{xy} &= \Sigma_{xy}^{\text{id}} + \Sigma_{xy}^{\text{ex}} \\ &= -\frac{m}{L^2} \sum_{i=1}^N v_{xi} v_{yi} - \frac{1}{L^2} \sum_{i=1}^{N-1} \sum_{j=i+1}^N x_{ij} F_{yij} \end{aligned} \quad (3)$$

where Σ_{xy}^{id} and Σ_{xy}^{ex} are the ideal gas stress and excess stress from particle interactions. Meanwhile, the shear viscosity can be calculated from

$$\eta = \frac{\Sigma_{xy}}{\dot{\gamma}} = \frac{\Sigma_{xy}^{\text{id}}}{\dot{\gamma}} + \frac{\Sigma_{xy}^{\text{ex}}}{\dot{\gamma}} = \eta^{\text{id}} + \eta^{\text{ex}} \quad (4)$$

We varied the number of particles N from 256 to 4096 to

verify that our results do not show significant system size dependence. Here we only show results for 1024.

3 Structural and dynamical signatures

Figure 1 shows an example of the flow curve (i.e., the shear-rate dependence of the shear stress or viscosity) calculated at $\xi = 0.1$. The shear stress decreased when decreasing the shear rate and approached a plateau at low shear rates, indicating that the shear stress in the $\dot{\gamma} \rightarrow 0$ limit (i.e., the yield stress) was nonzero, as marked by the arrow in Figure 1.

At low shear rates, the flow curve could be well fitted with the Hershel-Buckley formula, $\Sigma_{xy} = \Sigma_y + A\dot{\gamma}^\nu$, where Σ_y is the yield stress, and A and ν are fitting parameters. For the case shown in Figure 1, $\nu \approx 0.53$, consistent with the $1/2$ observed in colloidal experiments [15]. Due to the existence of the yield stress, $\eta = \Sigma_{xy} / \dot{\gamma} = \Sigma_y \dot{\gamma}^{-1} + A\dot{\gamma}^{\nu-1} \approx \Sigma_y / \dot{\gamma}$ at low shear rates. Apparently the viscosity decreases as the shear rate is increased (Figure 1) until a crossover shear rate $\dot{\gamma}_c$ is achieved; higher than this, the Hershel-Buckley formula no longer works and the viscosity increases with shear rate instead. Therefore, $\dot{\gamma}_c$ marks the transition from the shear thinning flow to the shear thickening flow. As will be discussed below, our shear thickening flows are associated with the high-temperature and high-pressure gaslike states induced by very high shear rates, which are in the same regime as previous simulations [9–12] but in a different re-

gime from recently reported DST flows that arose from the unjamming-jamming transition at low shear rates [16–18].

To check if our results were due to the Lees-Edwards boundary conditions, we studied systems confined between two rough walls at constant volume with the top wall moving in the x direction at a constant speed and observed similar results. It is thus interesting to ask if any experimentally accessible signatures are associated with the shear thinning-thickening transition observed here.

Figure 2(a–c) shows the snapshots of the shear flows at different shear rates. The system undergoes apparent structural changes from shear thinning to shear thickening. In the shear-thinning regime (Figure 2(a)), the system is roughly

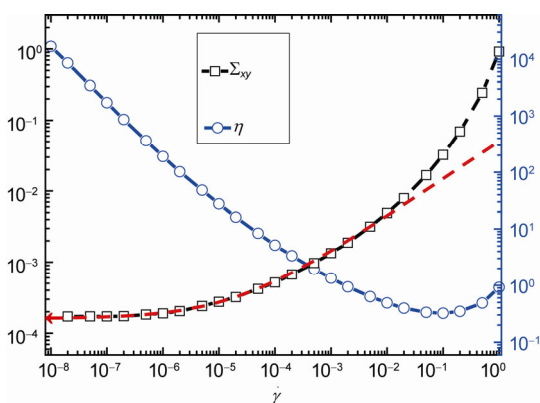


Figure 1 Shear-rate dependence of the shear stress Σ_{xy} and viscosity η with the damping coefficient $\xi=0.1$. The dashed line is the fitting to the low shear-rate flow curve with the Hershel-Buckley formula: $\Sigma_{xy} = 1.6 \times 10^{-4} + 0.05 \times \dot{\gamma}^{0.53}$. The arrow points to the yield stress $\Sigma_y \approx 1.6 \times 10^{-4}$.

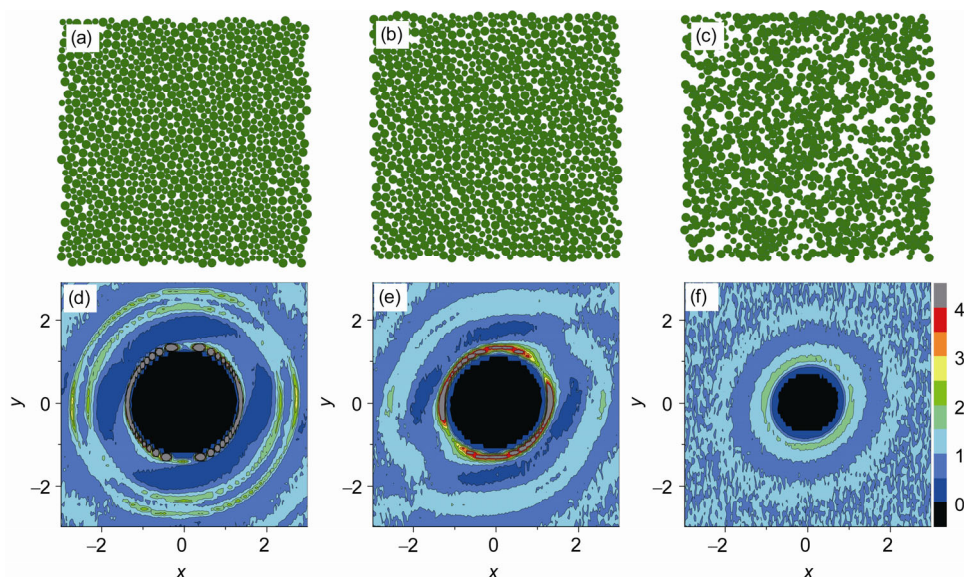


Figure 2 (a–c) Snapshots of shear flows at $\xi=0.1$ and $\dot{\gamma}=0.01, 0.1$, and 1 with $\dot{\gamma}_c \approx 0.1$; (d–f) pair distribution function $g(x, y)$ for systems shown in (a–c). The color bar quantifies the value of $g(x, y)$ (color online).

uniform in space; whereas in the shear-thickening regime (Figure 2(c)), particles tend to form instantaneous clusters with large overlaps and leave behind more voids. We were therefore inspired to search for possible structural signatures of the shear thinning-thickening transition.

Recent studies have attempted to bridge shear rheology and microstructure [13,33–35], although it has also been argued that there is no necessary connection between microstructure and the occurrence of shear thickening [5]. To characterize structure, we calculated the pair distribution function for large particles, $g(\vec{r}) = g(x, y) = \frac{L^2}{(N/2)^2} \left\langle \sum_i \sum_{j \neq i} \delta(\vec{r}(x, y) - \vec{r}_{ij}) \right\rangle$, with special attention to its

anisotropy. The sums are over all large particles and $\langle \cdot \rangle$ denotes the time average. The contour plots of $g(\vec{r})$ shown in Figure 2(d–f) indicate that $g(\vec{r})$ is asymmetric in the vicinity of the shear thinning-thickening transition; the contours are elongated in the direction of $y=x$ and compressed in the perpendicular direction.

In order to quantify the asymmetry, we show in the insets of Figure 3 the pair distribution functions parallel and perpendicular to $y=x$, respectively denoted as $g^{\parallel}(r)$ and $g^{\perp}(r)$. The first peak of $g^{\perp}(r)$ is higher than that of $g^{\parallel}(r)$. Moreover, $r_s^{\perp} < r_s^{\parallel}$, where $r_s^{\perp}$ and $r_s^{\parallel}$ denote the lengths at which $g^{\perp}(r)$ and $g^{\parallel}(r)$ start to be greater than zero. These features imply that when the structure is asymmetric, particles tend to have larger and more uniform overlaps in the direction perpendicular to $y=x$. We therefore define $\alpha = r_s^{\parallel} / r_s^{\perp}$ to characterize the asymmetry. Figure 3 shows that with the increase of the shear rate α first increases and then drops after a maximum. The maximum emerges approximately at $\dot{\gamma}_c$. The shear thinning-

thickening transition is thus signified by the most pronounced structural anisotropy.

Structural changes may also induce dynamical changes at the shear thinning-thickening transition. To capture these possible changes, we calculated the y component of the mean squared displacement (MSD) for large particles, $\langle \Delta y^2 \rangle$, where $\langle \cdot \rangle$ denotes the particle and ensemble average. As shown in the inset of Figure 4(a), there is a ballistic motion at short times ($\langle \Delta y^2 \rangle \sim t^2$) followed by diffusion at long times ($\langle \Delta y^2 \rangle \sim t$). Apparently, the relaxation time τ decreases as the shear rate increases. Here, τ is determined from $\langle \Delta y^2(\tau) \rangle \sim 10\sigma^2$.

When we plotted the MSDs against t/τ , as shown in Figure 4(a), the MSDs in the shear thinning regime collapsed nicely onto the same curve and the ballistic parts of the shear thickening curves remained apart. The scaling col-

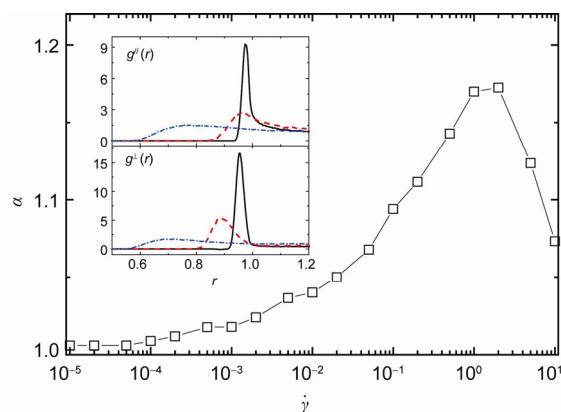


Figure 3 Anisotropy of $g(\vec{r})$ evaluated by α at various shear rates for shear flows at $\xi=0.1$. Inset: Pair distribution functions parallel and perpendicular to $y=x$, $g^{\parallel}(r)$ and $g^{\perp}(r)$, at $\dot{\gamma}=0.01$ (solid), 0.1 (dashed), and 1 (dot-dashed).

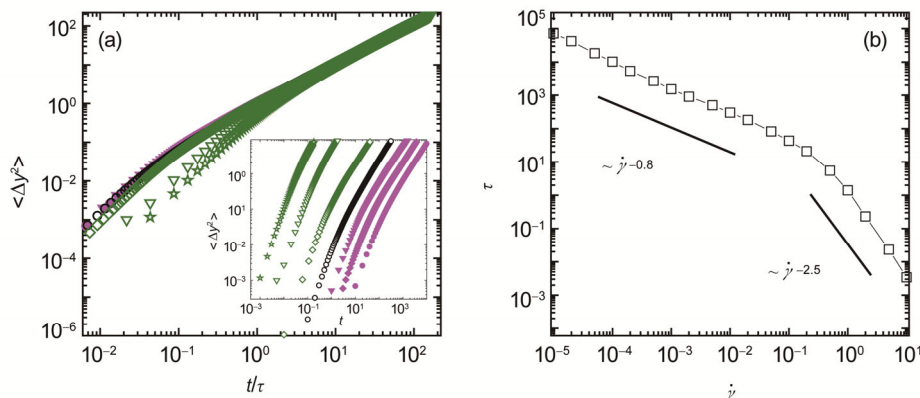


Figure 4 (a) Mean squared displacement $\langle \Delta y^2 \rangle$ in the shear-thinning (magenta solid symbols) and -thickening (green empty symbols) regimes at $\xi=0.1$. The black circles are at $\dot{\gamma} \approx \dot{\gamma}_c$. The shear rate increases from the right to the left in the inset. In the main panel, t is scaled by the relaxation time τ ; (b) shear-rate dependence of the relaxation time τ at $\xi=0.1$. The solid lines show power law behaviors in the shear-thinning and -thickening regimes.

lapse of the shear-thinning curves implies that particles move ballistically to approximately the same distance before diffusing. In the shear-thickening regime, however, particles move ballistically to a longer distance as the shear rate is increased. As shown in Figure 2(c), particles form instantaneous clusters in shear-thickening flows and leave many voids. If clustered particles move collectively at short times, the voids allow them to move ballistically to a longer distance, a process that may depict the short-time dynamics of shear-thickening flows. Further analysis of instantaneous clusters is necessary to verify our conjecture and gain a better understanding of shear thickening.

The distinct dynamics between shear-thinning and -thickening flows can also be identified from the shear-rate dependence of the relaxation time. As shown in Figure 4(b), $\tau \sim \dot{\gamma}^{-0.8}$ in the shear-thinning regime, in agreement with a recent experiment [36], whereas $\tau \sim \dot{\gamma}^{-2.5}$ in the shear-thickening regime. The distinct shear-rate scaling of the relaxation time, together with the breakdown of the scaling collapse of the short-time MSDs, thus serve as dynamical signatures of the shear thinning-thickening transition.

4 Decomposition of the shear viscosity

Thus far we have revealed the structural and dynamical signatures of the shear thinning-thickening transition for our model systems, which are accessible to experiments on colloidal suspensions or granular materials. Next we show another underlying condition for shear thickening to occur, which can be obtained from the decomposition of the shear viscosity based on the consideration of the energy balance.

In a steady shear flow, energy injected into the flow by the shear force is dissipated by the damping force, from which we obtain for our model:

$$\Sigma_{xy} \dot{\gamma} L^2 = \xi m \sum_{(i,j)} \bar{v}_{ij}^2 \quad (5)$$

where the sum is over all interacting particle pairs. For the model systems studied here, simple calculations of Eq. (5) lead to the eddy viscosity [9].

$$\eta_e = \eta_{-2}^T + \eta_{-2}^C + \eta_{-1} + \eta_0 \quad (6)$$

where

$$\eta_{-2}^T = \frac{\xi m}{L^2} \sum_{(i,j)} (\bar{v}_i^2 + \bar{v}_j^2) \dot{\gamma}^{-2} \approx \frac{2\xi NZ_b}{L^2} k_B T_g \dot{\gamma}^{-2} \quad (7)$$

$$\eta_{-2}^C = -\frac{2\xi m}{L^2} \sum_{(i,j)} (\bar{v}_i \cdot \bar{v}_j) \dot{\gamma}^{-2} \quad (8)$$

$$\eta_{-1} = \frac{2\xi m}{L^2} \sum_{(i,j)} (y_{ij} v_{xij}) \dot{\gamma}^{-1} \quad (9)$$

$$\eta_0 = \frac{\xi m}{L^2} \sum_{(i,j)} y_{ij}^2 \quad (10)$$

On the right-hand side of Eq. (7), z_b is the average coordination number and $T_g = \frac{m}{2Nk_B} \sum_{i=1}^N \bar{v}^2$ denotes the kinetic or granular temperature (i.e., effective temperature at short time scales) [28,37]. When introducing T_g in Eq. (7), we assume that the random part of the kinetic energy, $\sum_{i=1}^N \frac{1}{2} m \bar{v}^2$, is equally partitioned to all particles.

The decomposition of the shear viscosity by Eqs. (4) and (6) enables us to find the actual source of the shear stress in control of the rheology. Eq. (4) is the microscopic expression of the viscosity from virial theorem, whereas Eq. (6) is straightforwardly derived from the energy balance. In homogeneous steady flows, we would expect that $\eta = \eta_e$.

Figure 5(a) shows flow curves at four different damping coefficients ranging from 0.1 to 1, which all exhibit the shear thinning-thickening transition at the damping coefficient dependent crossover shear rate $\dot{\gamma}_c(\xi)$. When scaling $\dot{\gamma}$ by ξ , we notice that $\dot{\gamma}_c \sim \xi$. Scaling collapse at high shear rates in the shear thickening regime is also observed when η/ξ is plotted against $\dot{\gamma}/\xi$, which approximately obeys a power law scaling $\eta \sim \dot{\gamma}^2$. This scaling is different from the Bagnoldian behavior ($\eta \sim \dot{\gamma}$) [16–18,38], which arises from rigid particle contacts. Empirically, shear thickening with the change of the viscosity with respect to the shear rate faster than Bagnoldian can be treated as DST [5]. Apparently, the DST observed in our model systems should result from soft particle interactions and is consistent with the increased particle overlaps at high shear rates as discussed above, which make it different from the DST induced by the jamming transition of frictional particles.

As compared in Figure 5(a), $\eta \approx \eta_e$ as expected, except in the vicinity of $\dot{\gamma}_c$ where $\Delta\eta = \eta_e - \eta > 0$. $\Delta\eta$ tends to increase with ξ . We verified that $\Delta\eta > 0$ is not a transient behavior by observing no time evolution. In the shear-rate regime where $\Delta\eta$ is apparently greater than 0, the kinetics show visible anisotropy (e.g., $\sum_i v_{xi}^2 > \sum_i v_{yi}^2$), which is consistent with the emergence of the structure anisotropy in the same shear-rate regime discussed above. The anisotropy should thus account for the nonzero $\Delta\eta$. Although η and η_e are not completely equal, they exhibit the same $\dot{\gamma}_c$; therefore, this inequality does not affect our respective discussions about the shear thinning-thickening transition using Eqs. (4) and (6).

In Figure 5(b), we compare η^{id} with η^{ex} . At low shear rates, the random motion of particles is so slow that

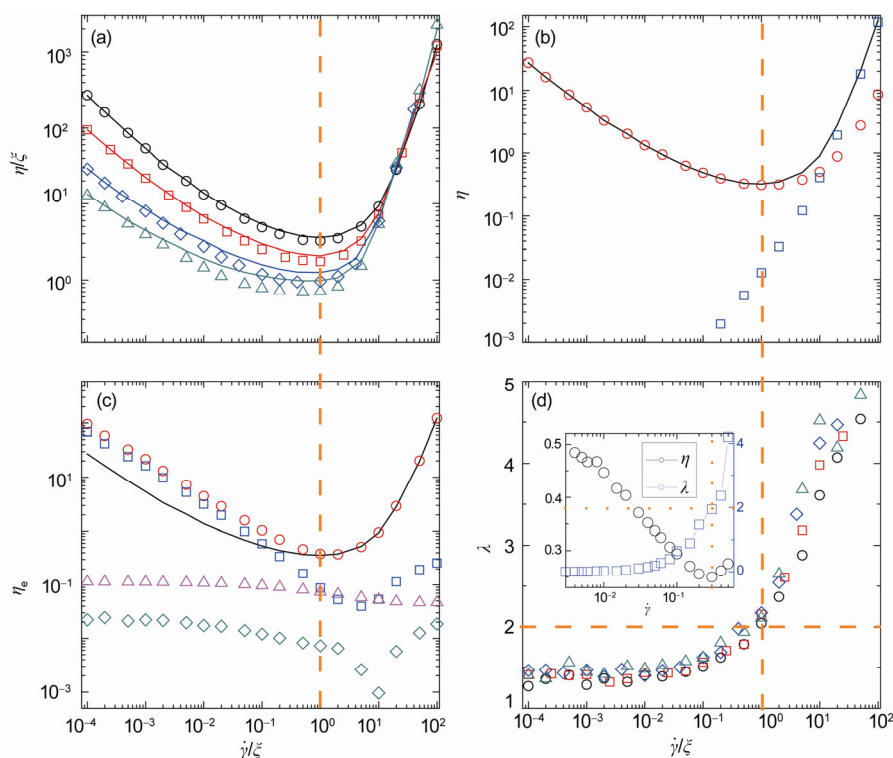


Figure 5 (a) Viscosity η calculated from Eq. (4) against the shear rate $\dot{\gamma}$ at $\xi=0.1$ (black circles), 0.2 (red squares), 0.5 (blue diamonds), and 1.0 (green triangles). The solid curves are eddy viscosity η_e calculated from Eq. (6); (b) shear-rate dependence of the ideal gas viscosity η^{id} (blue squares) and excess viscosity η^{ex} (red circles) at $\xi=0.1$. The black curve is the total viscosity; (c) shear-rate dependence of the four components of the eddy viscosity, η_{-2}^T (red circles), $-\eta_{-2}^C$ (blue squares), $|\eta_{-1}|$ (green diamonds), and η_0 (magenta triangles) at $\xi=0.1$. The solid curve is the total eddy viscosity; (d) shear-rate dependence of the rate of granular temperature increase λ at the same damping coefficients as (a). Inset of (d): comparison of η and λ for a thermostated system at $\frac{m}{Nk_B} \sum_i v_{yi}^2 = 0.1$.

$\eta^{\text{ex}} \gg \eta^{\text{id}}$. With increasing shear rate, η^{id} grows and eventually beats η^{ex} . $\eta^{\text{id}} = \eta^{\text{ex}}$ at a crossover shear rate $\dot{\gamma}_g$. When $\dot{\gamma} > \dot{\gamma}_g$, the system behaves effectively as a high-temperature and high-pressure gas, with ξ^{-1} and $\dot{\gamma}_c^{-1}$ as the dominant time scales; this may be the cause of the high shear-rate scaling collapse shown in Figure 5(a). η^{ex} also exhibits the shear thinning-thickening transition lagging slightly behind $\dot{\gamma}_c$. The shear thinning-thickening transition observed here is thus determined by the competition between the shear thinning-thickening transition in η^{ex} and the growth rate of η^{id} with the shear rate.

In contrast to Eq. (4), the decomposition of the shear viscosity using Eq. (6) is more useful to reveal an underlying thermodynamic-like condition for shear thickening to occur. In Figure 5(c), we show all the four viscosity components in Eq. (6). In the whole shear-rate regime studied here, η_0 and η_{-1} are small and negligible, $\eta_{-2}^T > 0$, and $\eta_{-2}^C < 0$. At low shear rates, $|\eta_{-2}^C|$ is smaller than but

comparable to η_{-2}^T , and shows similar shear-rate dependence to η_{-2}^T . Near $\dot{\gamma}_c$, however, η_{-2}^T becomes significantly larger than $|\eta_{-2}^C|$. Shear thickening is therefore mainly determined by η_{-2}^T . Assuming that the average coordination number z_b does not vary largely with the shear rate, which is actually true for our systems, Eq. (7) implies that if the kinetic temperature T_g varies more steeply than $\dot{\gamma}^2$, shear thickening would occur.

In athermal shear flows with a constant ξ , T_g increases with the shear rate. To estimate the rate of the T_g increase, we defined a quantity $\lambda = \frac{d(\ln T_g)}{d(\ln \dot{\gamma})}$ and plotted it against

the shear rate (Figure 5(d)). At low shear rates, $\lambda \approx 1.5$, which indicates that $T_g \sim \dot{\gamma}^{1.5}$. This scaling law stops working near $\dot{\gamma}_c$; at this point, when $\dot{\gamma} > \dot{\gamma}_c$ and $\lambda > 2$, shear thickening happens—exactly as expected from our analysis of Eq. (6).

Having established a direct link between the kinetic temperature and shear thickening for our model systems, we then asked if this link was specific to our systems. First of all, we have verified that systems with harmonic, Hertzian, and Lennard-Jones interactions came to the same conclusion. Next, we tried to find out if $\lambda > 2$ was also the condition of shear thickening in other typical model systems. Other than our athermal model, typical models for the study of shear flows include the foam model [39–41], Langevin dynamics [42,43], and thermostated shear flows [7,9–12,37].

For the foam model described by the equation of motion, $\zeta \vec{v}_i = \sum_j \vec{F}_{ij}^e$, where ζ is equivalent to ξm in our

model, simple calculations resulted in $\eta_e \sim \sum_i \vec{v}_i^2 \dot{\gamma}^{-2}$. Thus

we can immediately see that shear thickening happens when $\frac{d(\ln \sum_i \vec{v}_i^2)}{d(\ln \dot{\gamma})} > 2$, equivalent to $\lambda > 2$ except that particle

inertia is ignored. When inertia is included, $\lambda > 2$ can be straightforwardly obtained as well.

For Langevin dynamics, a random force $\vec{R}_i(t)$ should be added to the equation of motion for the foam model. Because $\vec{v}_i$ and $\vec{R}_i$ are uncorrelated [43], we would expect the same condition for shear thickening to occur.

For thermostated shear flows, we already knew that when T_g is fixed shear thickening cannot happen [7], whether or not it is an artifact of thermostat [26]. Even with more realistic thermostat, it has been reported that T_g is much higher than the bath temperature [26]. These observations may indicate that shear thickening is also related to the kinetic temperature in thermostated systems. In the inset of Figure 5(d), we show both η and λ against the shear rate for a thermostated shear flow governed by

$$\frac{d\vec{v}_i}{dt} = \frac{1}{m} \sum_j \vec{F}_{ij}^e - \dot{\gamma} v_{yi} \hat{x} - \alpha_T v_{yi} \hat{y}, \text{ where } \alpha_T \text{ maintains a}$$

constant kinetic energy in the y direction [37] but leaves the kinetic energy in the x direction and T_g unconstrained. For this model, energy balance resulted in $\eta_e \sim \alpha_T \sum_i v_{yi}^2 \dot{\gamma}^{-2}$,

from which $\lambda > 2$ cannot be recognized. Interestingly, $\lambda > 2$ still signifies shear thickening for the case shown here.

In this paper, we mainly concentrate on the athermal model described by Eqs. (1) and (2). In order to verify whether $\lambda > 2$ is a robust condition of shear thickening for thermostated systems, intensive studies at various bath temperatures and densities and with different interaction potentials and thermostats would be required.

For foam model and Langevin dynamics, $\lambda > 2$ is a straightforward consequence of the energy balance required for shear thickening to occur. For our athermal model, this consequence is not obvious because the precondition is that η_{-2}^T is significantly larger than the other three terms in Eq.

(6), which turns out to be true. For thermostated shear flows, $\lambda > 2$ is totally unpredictable. Our results and analysis suggest that the occurrence of shear thickening is strongly coupled with the behavior of the kinetic energy, regardless of models. Therefore, $\lambda > 2$ may be a robust thermodynamic-like condition for shear thickening to occur.

5 Conclusions

In steady shear flows of dense non-Brownian systems consisting of soft particles, we observed the shear thinning-thickening transition. This transition was signified by the strongest structural anisotropy and apparent changes in the scaling of dynamical quantities such as mean squared displacement and relaxation time. The shear-thickening flows observed here are high-temperature and high-pressure gas-like states with large particle overlaps, which should be impossible in hard-sphere systems that prohibit particle overlaps.

Note that our systems are jammed solids at rest, whereas in most of the colloidal experiments the packing fraction is below ϕ_c , the critical packing fraction of the jamming transition. Therefore, the shear thickening observed here must have been induced by a distinct mechanism from the unjamming-jamming transition proposed for unjammed systems [5]. Because we did not include hydrodynamics in our model, the lubrication force [2] is not essential either. Furthermore, jammed solids are disordered, which may already exclude the order-disorder transition [20] as the underlying cause of shear thickening. Although the shear thickening observed here cannot be simply explained by any of the three popular theories discussed above, it may be too early to proclaim the existence of new mechanisms of shear thickening based simply on our observations. After all, our systems are in different density regimes from those that are commonly concerned. Whether shear thickening can happen in experimental jammed systems composed of deformable particles such as PNIPAM colloidal systems is thus the key to verifying our major findings. For shear thickening to be evident, the particles should be deformable with hard cores, in order to cause particle overlap, and should also supply sufficiently large confining stress to ensure visible shear thickening [5].

Jammed solids have nonzero yield stress, which boosts shear thinning at low shear rates and impedes the occurrence of shear thickening. In shear flows of dilute colloidal suspensions, a Newtonian regime can sometimes be observed after shear thinning [2,44], which was unsurprisingly absent in our systems. When the yield stress is turned on for unjammed systems (for instance, by adding attraction) and is small, a Newtonian regime can occur after shear thinning regime at low shear rates. The Newtonian regime shrinks as the strength of the attraction increases and eventually dis-

appears when the yield stress is so large that the shear-thinning regime directly encounters the shear-thickening regime. What we have observed in our model systems should therefore emerge, although the packing fraction is low; this hypothesis, of course, requires further investigation.

Furthermore, by applying a simple analysis of the shear viscosity based on the energy balance, we find an underlying thermodynamic-like condition for shear thickening to occur: $d(\ln T_g)/d(\ln \dot{\gamma}) > 2$. We suggest that this condition may be robust to various model systems, which still needs to be verified especially for systems in which the condition cannot be straightforwardly derived. In any case, this condition may provide us with a possible reference to manipulate the occurrence of shear thickening by tuning the properties of constituent particles.

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